

CROOKSTON, MN, USA

WMO: 727452

Lat: 47.842N

Lon: 96.621W

Elev: 274

StdP: 98.08

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04912

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.9	-27.6	-34.1	0.2	-30.5	-31.3	0.2	-27.1	15.7	-14.9	14.5	-13.6	3.1	310	0.671

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	31.0	22.4	28.9	21.2	27.6	20.0	23.9	28.5	22.6	27.4	21.5	26.2	6.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	17.6	27.4	21.2	16.3	26.1	19.5	14.7	24.5	72.8	28.7	68.0	27.5	63.6	26.4	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.8	11.5	10.3	DB	-33.0	34.1	3.8	3.7	-35.8	36.8	-38.0	39.0	-40.1	41.0	-42.8	43.8	(4)
(5)			WB	-33.2	25.9	3.6	1.2	-35.8	26.8	-37.9	27.4	-40.0	28.1	-42.6	29.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.6	-14.2	-12.1	-4.1	5.1	12.4	18.2	20.4	19.3	14.7	6.8	-2.3	-10.3	(6)	
	DBStd	13.58	8.23	7.70	7.63	5.65	5.02	3.67	3.11	3.30	4.73	5.10	6.47	7.22	(7)	
	HDD10.0	3143	750	619	439	163	33	1	0	0	11	127	371	631	(8)	
	HDD18.3	5231	1008	852	696	396	193	47	15	28	128	359	621	889	(9)	
	CDD10.0	1162	0	0	2	17	107	245	324	288	152	27	1	0	(10)	
	CDD18.3	208	0	0	0	0	8	41	81	57	19	1	0	0	(11)	
	CDH23.3	2038	0	0	0	7	125	397	739	555	198	17	0	0	(12)	
	CDH26.7	529	0	0	0	1	33	97	196	147	52	3	0	0	(13)	
(14) Wind	WSAvg	4.6	5.0	4.7	4.8	4.9	5.0	4.1	3.5	3.7	4.4	4.9	4.8	4.8	(14)	
(15) Precipitation	PrecAvg	579	13	13	19	32	71	105	84	81	64	52	23	19	(15)	
	PrecMax	701	32	34	40	76	179	187	173	224	146	139	101	43	(16)	
	PrecMin	404	1	1	3	11	15	25	17	14	12	1	0	2	(17)	
	PrecStd	88	9	9	12	18	40	45	41	46	30	34	24	10	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.9	6.3	17.5	23.7	31.0	32.2	32.4	32.5	31.3	26.0	16.1	4.3	(19)	
		MCWB	1.4	4.0	12.2	13.4	19.3	23.0	23.7	23.4	21.5	16.4	10.4	1.9	(20)	
	2%	DB	1.2	2.8	12.1	20.9	27.0	29.0	30.8	30.0	27.8	21.3	12.0	2.4	(21)	
		MCWB	0.4	1.2	8.9	11.9	16.5	20.8	23.5	21.6	19.2	14.5	8.3	1.0	(22)	
	5%	DB	-1.0	1.3	7.8	17.5	23.8	27.4	28.7	27.9	26.0	17.6	8.7	1.0	(23)	
		MCWB	-1.4	0.3	5.6	10.2	14.7	19.1	22.2	20.4	18.1	12.1	6.1	0.0	(24)	
	10%	DB	-2.6	-1.0	5.0	14.2	21.6	26.0	27.4	27.0	22.8	14.9	6.2	-1.1	(25)	
		MCWB	-3.4	-1.5	3.1	8.5	13.9	18.5	20.9	19.7	16.9	10.5	3.8	-1.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.8	4.5	13.0	14.9	20.1	24.8	25.6	24.7	23.0	17.6	11.2	2.9	(27)	
		MCDB	2.7	6.3	15.3	21.1	26.5	29.1	30.4	29.7	28.4	23.3	14.8	3.9	(28)	
	2%	WB	0.3	1.7	8.7	12.3	18.3	22.5	24.3	23.1	20.9	15.2	8.3	0.9	(29)	
		MCDB	1.1	2.9	11.5	18.7	23.8	27.5	28.7	28.1	25.7	19.8	10.7	2.1	(30)	
	5%	WB	-1.4	0.3	5.8	10.7	16.5	20.7	23.1	21.8	19.2	12.8	6.1	-0.2	(31)	
		MCDB	-1.1	1.2	8.0	16.6	21.5	25.5	27.4	26.6	23.7	17.2	8.6	0.6	(32)	
	10%	WB	-3.4	-1.7	3.3	9.2	14.9	19.4	21.8	20.5	17.8	10.8	3.9	-1.5	(33)	
		MCDB	-2.6	-1.2	5.0	14.2	20.0	24.0	26.3	24.9	22.0	14.3	6.1	-1.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.0	9.5	9.0	10.7	12.2	11.0	11.5	12.4	11.9	9.8	8.3	8.1	(35)	
		MCDBR	9.0	8.6	11.5	16.1	15.9	13.3	13.1	14.0	14.8	14.4	11.9	9.1	(36)	
	5% WB	MCWBR	8.4	7.6	8.7	9.4	8.1	6.8	7.3	7.2	7.9	8.7	8.4	7.9	(37)	
		MCWBR	9.0	8.4	11.3	14.8	12.9	11.3	11.7	12.2	12.4	13.5	11.2	8.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.292	0.311	0.304	0.357	0.373	0.388	0.389	0.391	0.356	0.314	0.284	0.277	(40)	
		taud	2.238	2.205	2.420	2.341	2.357	2.343	2.343	2.341	2.432	2.524	2.500	2.358	(41)	
	Ebn at Noon		769	838	921	895	889	875	868	850	851	843	801	754	(42)	
		Edh at Noon	78	101	96	116	118	121	119	115	95	74	61	62	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.59	2.79	3.91	4.88	5.49	6.01	6.35	5.30	3.90	2.42	1.43	1.10	(44)	
	RadStd		0.17	0.26	0.41	0.44	0.35	0.45	0.27	0.37	0.28	0.35	0.17	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+89	N/A
(47) Regional (25 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page